

[Homework 4] Poisson & Brownian Motion

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Problem 1

- The goal is, the first customer arrived after T-s is the last customer. The goal fails when more than 1 customers arrive in s hours. If no customer arrives, Joe will leave at T. I doubt whether the goal should be considered reached in this case, so I will consider both. The Poisson distribution with λ is:

$$\Pr[X = k] = \frac{\lambda^k}{k!} e^{-\lambda} \cdot \text{sign } k \quad (1)$$

and we know we can add the λ s when considering the sum of multiple variables that satisfy $\text{Pois}(\lambda)$. Then $\Pr[X_s = 1] = s\lambda e^{-s\lambda}$, $\Pr[X_s = 0] = e^{-s\lambda}$.

Let's call the case where there must be one customer case 1, and where there can be no customer case 2.

The answer for case 1 is $s\lambda e^{-s\lambda}$ and the answer for case 2 is $(1 + s\lambda)e^{-s\lambda}$.

- To find the optimal s, differentiate the answer in the last question with respect to s:
Case 1: $\frac{d(s\lambda e^{-s\lambda})}{ds} = \lambda e^{-s\lambda} - s\lambda^2 e^{-s\lambda} = 0$, $s = \frac{1}{\lambda}$
Case 2: $\frac{d((1+s\lambda)e^{-s\lambda})}{ds} = \lambda e^{-s\lambda} - s\lambda^2 e^{-s\lambda} - \lambda e^{-s\lambda} = 0$, err, I guess case 1 is correct.

Problem 2

- To prove $\Pr[X = \lambda + k] \geq \Pr[X = \lambda - k - 1]$, that is to prove:

$$\frac{\lambda^{\lambda+k}}{(\lambda+k)!} e^{-\lambda} \geq \frac{\lambda^{\lambda-k-1}}{(\lambda-k-1)!} e^{-\lambda} \quad (2)$$

Simplify,

$$\frac{\lambda^{2k+1}(\lambda-k-1)!}{(\lambda+k)!} = \frac{\lambda^{2k+1}}{\prod_{i=-k}^k (\lambda+i)!} \geq 1 \quad (3)$$

$$\text{LHS} = \frac{\lambda^{2k}}{\prod_{i=1}^k (\lambda^2 - i^2)} \geq 1 \quad (4)$$

Equation 4 is obvious. Then

$$\sum_{k=0}^{\lambda-1} \Pr[X = k] \leq \sum_{k=0}^{\lambda-1} \Pr[X = \lambda + k] \leq \sum_{k=0}^{\infty} \Pr[X = \lambda + k] = \Pr[X \geq \lambda] \quad (5)$$

$$\Pr[X \geq \lambda] \geq \frac{1}{2} \quad (6)$$

- For this problem, $\sum_{k=1}^n Y_k \sim \text{Poisson}(m\lambda)$.

$$\begin{aligned} \mathbf{E}[f(Y_1, Y_2, \dots, Y_n)] &= \sum_{k=0}^{\infty} \mathbf{E}\left[f(Y_1, Y_2, \dots, Y_n) \mid \sum_{i=1}^n Y_i = k\right] \Pr\left[\sum_{i=1}^n Y_i = k\right] \\ &\geq \sum_{k=\lambda}^{\infty} \mathbf{E}\left[f(Y_1, Y_2, \dots, Y_n) \mid \sum_{i=1}^n Y_i = m\right] \Pr\left[\sum_{i=1}^n Y_i = m\right] \\ &= \mathbf{E}[f(X_1, X_2, \dots, X_n)] \sum_{k=\lambda}^{\infty} \Pr\left[\sum_{i=1}^n Y_i = m\right]. \end{aligned} \quad (7)$$

$$\sum_{k=1}^n Y_k \sim \text{Poisson}(m\lambda), \sum_{j=\lambda}^{\infty} \mathbf{Pr}[\sum_{k=1}^n Y_k = j] \geq \frac{1}{2}$$

$$2\mathbf{E}[f(Y_1, Y_2, \dots, Y_n)] \geq \mathbf{E}[f(X_1, X_2, \dots, X_n)]. \quad (8)$$

- Use the corollary we get in the last question: the case there exists five students who share the same birthday can be describes as

$$\max(X_1, \dots, X_m) \geq 5 \quad (9)$$

where X_i is the number of students whose birthday is i . Then we let $\sum_{i=1}^m Y_i = n$ and $Y_i \sim \text{Pois}(\lambda = \frac{n}{m})$, Let

$$f(X_1, X_2, \dots, X_m) = \mathbf{1}[X \geq 5] = \mathbf{1}[\max(X_1, \dots, X_m) \geq 5] \quad (10)$$

$$\begin{aligned} \mathbf{Pr}[\max(X_1, \dots, X_m) \geq 5] &= \mathbf{E}[f(X_1, X_2, \dots, X_m)] \\ &\leq 2\mathbf{E}[f(Y_1, Y_2, \dots, Y_m)] \\ &= 2\mathbf{Pr}[\max(Y_1, \dots, Y_m) \geq 5] \\ &= 2 - 2\mathbf{Pr}[\max(Y_1, \dots, Y_m) < 5] \\ &= 2 - 2\left(\prod_{i=1}^m \mathbf{Pr}[Y_i < 5]\right) \\ &\leq 2 - 2(0.999987)^{365} \approx 0.00947 \end{aligned} \quad (11)$$

0.999987 is because $\sum_{i=0}^4 \mathbf{Pr}[Y = i] \approx 0.9999870714$

Problem 3

- Prove according to definition: let's call $P(0) = c^{-\frac{1}{2}}W(ct)$
 1. $P(0) = c^{-\frac{1}{2}}W(0) = 0$
 2. $\forall 0 \leq t_1 \leq t_2, P(t_2) - P(t_1) = c^{-\frac{1}{2}}(W(ct_2) - W(ct_1))$ is mutually independent since $W(ct_2) - W(ct_1)$ is mutually independent
 3. $\forall s, t > 0, P(s+t) - P(s) \sim \mathcal{N}(0, c^{-1} \cdot ct) \rightarrow \mathcal{N}(0, t)$
 4. $W(t)$ is continuous almost surely. So is $P(t)$

Thus, $P(t)$ is a standard Brownian motion.

- Again,

1. $X(0) = 0$
2. $\forall 0 \leq t_1 \leq t_2, X(t_2) - X(t_1) = W(c+t_2) - W(c+t_1)$ is mutually independent
3. $\forall s, t > 0, X(s+t) - X(s) \sim \mathcal{N}(0, t)$
4. $W(t)$ is continuous almost surely. So is $X(t)$

$X(t)$ is a standard Brownian motion, and

$$\begin{aligned} \text{cov}(X(t_1), W(t_2)) &= \text{cov}(W(c+t_1) - W(c), W(t_2)) \\ &= \text{cov}(W(c+t_1), W(t_2)) - \text{cov}(W(c), W(t_2)) = 0 \end{aligned} \quad (12)$$

Thus it's independent.

- The original is equal to $\mathbf{Pr}[W(1) - W(\frac{1}{2}) > -W(\frac{1}{2}) \mid W(\frac{1}{2}) > 0]$, and according to the last question, $X(\frac{1}{2}) = W(\frac{1}{2} + \frac{1}{2}) - W(\frac{1}{2})$ is a standard Brownian motion independent of $W(t)$. And we let $W'(t) = -W(t)$. The new problem become $\mathbf{Pr}[X(\frac{1}{2}) - W'(\frac{1}{2}) > 0 \mid W'(\frac{1}{2}) < 0]$.

Use the total probability law,

$$\begin{aligned}
& \mathbf{Pr}[X(\frac{1}{2}) > W'(\frac{1}{2}) \mid W'(\frac{1}{2}) < 0] \\
&= \mathbf{Pr}[X(\frac{1}{2}) > W'(\frac{1}{2}) \mid W'(\frac{1}{2}) < 0, X(\frac{1}{2}) \geq 0] \mathbf{Pr}[X(\frac{1}{2}) \geq 0] + \\
& \quad \mathbf{Pr}[X(\frac{1}{2}) > W'(\frac{1}{2}) \mid W'(\frac{1}{2}) < 0, X(\frac{1}{2}) < 0] \mathbf{Pr}[X(\frac{1}{2}) < 0] \\
&= 1 \cdot \frac{1}{2} + \frac{1}{2} \cdot \frac{1}{2} \\
&= \frac{3}{4}
\end{aligned} \tag{13}$$

$\mathbf{Pr}[X(\frac{1}{2}) > W'(\frac{1}{2}) \mid W'(\frac{1}{2}) < 0, X(\frac{1}{2}) < 0] = \frac{1}{2}$ is because for two standard Brownian motion, all the situations are symmetric.

Problem 4

- To prove

$$\begin{aligned}
\mathbf{Pr}[X(t) \leq \delta] &= \mathbf{Pr}\left[\xi \leq \frac{\delta - \mu \cdot t}{\sigma\sqrt{t}}\right], \\
\mathbf{Pr}[\mu t + \sigma W(t) \leq \delta] &= \mathbf{Pr}\left[\xi \leq \frac{\delta - \mu \cdot t}{\sigma\sqrt{t}}\right],
\end{aligned} \tag{14}$$

Here $W(t) \sim \mathcal{N}(0, t)$. View $W(t)$ as $\sqrt{t}\xi$,

$$\mathbf{Pr}[\mu t + \sigma\sqrt{t}\xi \leq \delta] = \mathbf{Pr}\left[\xi \leq \frac{\delta - \mu \cdot t}{\sigma\sqrt{t}}\right], \tag{15}$$

This is much too obvious.

- According to the last question,

$$\begin{aligned}
\mathbf{Pr}[X(t) \geq 0] &= \mathbf{Pr}\left[\xi \geq \frac{-\mu \cdot t}{\sigma\sqrt{t}}\right], \\
\mathbf{Pr}[X(t) \leq \delta] &= \mathbf{Pr}\left[\xi \leq \frac{\delta - \mu \cdot t}{\sigma\sqrt{t}}\right],
\end{aligned} \tag{16}$$

Let a new ξ be $-\xi$ and replace it in the equations above:

$$\begin{aligned}
\mathbf{Pr}[X(t) \geq 0] &= \mathbf{Pr}\left[\xi \geq \frac{-\mu \cdot t}{\sigma\sqrt{t}}\right], \\
\mathbf{Pr}[0 \leq X(t) \leq \delta] &= \mathbf{Pr}\left[\frac{\mu \cdot t - \delta}{\sigma\sqrt{t}} \leq \xi \leq \frac{\mu \cdot t}{\sigma\sqrt{t}}\right], \\
\int_0^\infty \mathbf{Pr}[0 \leq X(t) \leq \delta] dt &= \int_0^\infty \mathbf{Pr}\left[\frac{\mu \cdot t - \delta}{\sigma\sqrt{t}} \leq \xi \leq \frac{\mu \cdot t}{\sigma\sqrt{t}}\right] dt
\end{aligned} \tag{17}$$

And the LHS is the definition of the expectation of T .

- This is to find $\mathbf{Pr}[0 \leq X(t) \leq \delta] = \mathbf{Pr}[f(0, \xi) \leq t \leq f(\delta, \xi)]$. $X(t) = \mu t + \sigma\sqrt{t}\xi$. Simply

we need to solve the quadratic equation.

$$\begin{aligned}
t &\geq \frac{\sigma^2 \xi^2}{\mu^2} \\
\sqrt{t} &\leq \frac{-\sigma \xi \pm \sqrt{\sigma^2 \xi^2 + 4\mu\delta}}{2\mu} \\
f(0, \xi) &= \frac{\sigma^2 \xi^2}{\mu^2} \\
f(\delta, \xi) &= \frac{\delta}{\mu} + \frac{\sigma^2 \xi^2}{2\mu^2} + \frac{\sigma \xi \sqrt{\sigma^2 \xi^2 + 4\mu\delta}}{2\mu^2}
\end{aligned} \tag{18}$$

- $\mathbf{E}f(\delta, \xi) = \mathbf{E}\left[\frac{\delta}{\mu} + \frac{\sigma^2 \xi^2}{2\mu^2} + \frac{\sigma \xi \sqrt{\sigma^2 \xi^2 + 4\mu\delta}}{2\mu^2}\right] = \frac{\delta}{\mu} + \frac{\sigma^2 \mathbf{E}[\xi^2]}{2\mu^2} + \frac{\sigma \mathbf{E}[\xi \sqrt{\sigma^2 \xi^2 + 4\mu\delta}]}{2\mu^2}$
 $\mathbf{E}[\xi^2] = 1, \mathbf{E}[\xi \sqrt{\sigma^2 \xi^2 + 4\mu\delta}] = 0$
 $\mathbf{E}[f(\delta, \xi)] = \frac{\delta}{\mu} + \frac{\sigma^2}{2\mu^2}$
- $\mathbf{E}[T] = \int_0^\infty \mathbf{Pr}\left[\frac{\mu \cdot t - \delta}{\sigma \sqrt{t}} \leq \xi \leq \frac{\mu \cdot t}{\sigma \sqrt{t}}\right] dt$
 $= \int_0^\infty \mathbf{Pr}[f(0, \xi) \leq t \leq f(\delta, \xi)] dt$
 $= \int_0^\infty \mathbf{Pr}[t \leq f(\delta, \xi)] dt - \int_0^\infty \mathbf{Pr}[t \leq f(0, \xi)] dt$
 $= \frac{\delta}{\mu} + \frac{\sigma^2}{2\mu^2} - \frac{\sigma^2}{2\mu^2} = \frac{\delta}{\mu}$

References

- [1] C. Zhang, Stochastic Processes (Spring 2023) [Online]. Available: <http://chihaozhang.com/teaching/SP2023/> [Accessed Mar. 24, 2023].